

Student 3 - Implementation Code Walkthrough

Slides 6–7 | Time: 2 min 30 sec | Role: Show the code, explain the logic

"Now let me walk you through the actual code. Six blocks — three on each slide."

| Click to Slide 05 — Implementation Part 1.

"Block one — the Data model."

| Point at Block 1 heading.

"The Item class looks like this — class Item, with int id, String name. The constructor uses this dot id equals id to set the value."

| Point at the code snippet on the left side of Block 1.

"On the right side — the diagram shows all eight fields. id, type, name, category, date, location, contact, status. Separated by pipes in the file."

| Point at the diagram on the right side showing the 8 fields.

"Block two — File storage. This is how we read and write data."

| Point at Block 2 heading.

"Each line in items dot txt looks like this — 1 pipe LOST pipe Wallet pipe Bags and so on."

| Point at the example line in the code.

"To read it, we split by pipe. String array p equals line dot split. Then we parse the ID with Integer dot parseInt p of 0."

| Point at the split and parseInt code lines.

"The diagram shows the result — p0 through p7. Eight fields from one line."

| Point at the diagram showing p0 p1 p2 etc.

"Block three — Validation. We check the date format using regex."

| Point at Block 3 heading.

"The regex is — backslash d 2, hyphen, backslash d 2, hyphen, backslash d 4. Two digits, hyphen, two digits, hyphen, four digits."

| Point at the regex in the code. Say each part slowly.

"Look at the examples. 15-09-2026 — accepted, shown in green. 9 slash 15 slash 26 — rejected, shown in red. The color tells you everything."

| Point at the green OK example, then the red REJECT example. Pause on each.

"Now part two — Search, Claim, and the Menu loop."

| Click to Slide 06 — Implementation Part 2.

"Block four — Search. This is where streams shine."

| Point at Block 4 heading.

"The code is just five lines. Items dot stream. Dot filter — check if the name contains the keyword. Dot collect to list. That is it."

| Point at each line of code as you say it.

"The execution flow on the right shows it visually. Items go in, stream, filter, collect — filtered list comes out. Clean pipeline."

| Point at the execution flow diagram. Trace the flow from left to right.

"Block five — Claim. This is a bit different."

| Point at Block 5 heading.

"We loop through all items. If the ID matches and status is not RESOLVED — we set status to RESOLVED. Then we rewrite the entire file."

| Point at the for loop, then the setStatus line, then the rewrite line.

"Why rewrite? Because every line in the file contains the status. When status changes, we must update everything."

| Make a circular motion to indicate "everything."

This is a key point — explain the why, not just the what.

"The execution flow shows it clearly. Before — id 7, status OPEN. After claim — id 7, status RESOLVED. Then rewrite items dot txt."

| Point at the before-after strip in the diagram.

"Block six — the Menu loop. This keeps the program running."

| Point at Block 6 heading.

"While true — the program never stops until you choose exit. It reads your input with Scanner, checks if it equals 1 for post lost, or 8 to break."

| Point at the while loop, then the if statements.

"The menu on the right shows all options — 1 Post Lost, 2 Post Found, 3 View Lost, 5 Search, 6 Claim, 8 Exit. User input goes in, action comes out."

| Point at the menu list in the execution flow.

"That is the code. Now let me hand over for the live demonstration — you will see it all running."

| Turn towards next speaker. Step back.